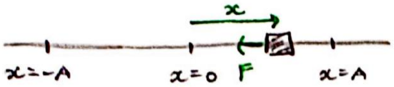
# Simple Harmonic Motion 1

## # Kinematics of SHM :




(Mean position) ⇒ where $\boxed{F_{net}=0}$

→ SHM occurs when restoring force is directly proportional to negative of displacement.

i.e $\boxed{\vec{F} \propto -\vec{x}}$

⇒ $F = -Kx$ $\quad \left[\omega = \sqrt{\frac{K}{m}}\right]$

$a = -\left(\frac{K}{m}\right)x \Rightarrow \boxed{a = -\omega^2 x}$

**Differential form of SHM :→**

Since, $a = -\omega^2 x$

⇒ $\boxed{\frac{d^2x}{dt^2} + \omega^2 x = 0}$

↳ Basic form, that should be satisfy by any eqn to be eqn of SHM

↳ Can have infinite solns, out of which functions like "sin" & "cos" are commonly used.

$\boxed{x = A\sin(\omega t + \phi)}$

$\boxed{x = Ae^{i\omega t}}$

**Time Period**

$\boxed{T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{K}}}$

**Velocity**

$\boxed{V(x) = \omega\sqrt{A^2 - x^2}}$ — $V_{max} = \pm\omega A,\ x=0$; $V_{min} = 0,\ x = \pm A$

$\boxed{V(t) = A\omega\cos\omega t}$

**Acceleration**

$\boxed{a(x) = -\omega^2 x}$ — $a_{max} = \pm\omega^2 A,\ x = \pm A$; $a_{min} = 0,\ x = 0$

$\boxed{a(t) = -\omega^2 A\sin\omega t}$

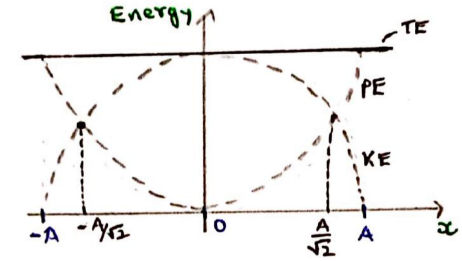
## # Energy of SHM :

1) $\boxed{KE = \frac{1}{2}m\omega^2(A^2 - x^2)}$

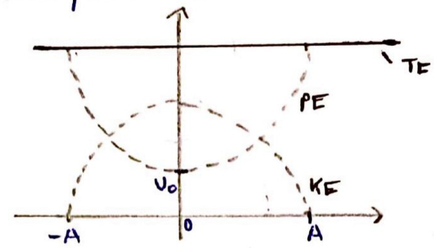
2) $\boxed{PE = \frac{1}{2}m\omega^2 x^2}$

3) $\boxed{TE = \frac{1}{2}m\omega^2 A^2}$ (constant)

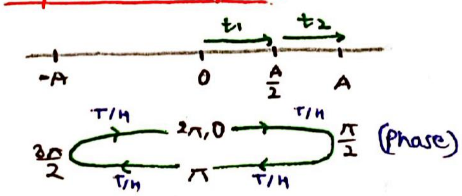
**General Graph :→**




→ If particle has some PE a $U_0$ at mean position ($x=0$), then




## # Important Analysis :




1) Time to go from mean to extreme or vice versa is same

i.e $\boxed{t_{0\to A} = T/4}$

2) $\boxed{t_1 = t_{0\to A/2} = T/12}$

3) $\boxed{t_3 = t_{\frac{A}{2}\to A} = T/6}$

## # Steps to find time Period :

**Linear SHM**

1) Give linear displ. $x$ from mean position

2) Find accn by $\boxed{F = ma}$

3) We get $\boxed{a = -(\text{const.})x}$ ↳ $\omega^2$

**Angular SHM**

1) Give angular disp. $\theta$ from mean

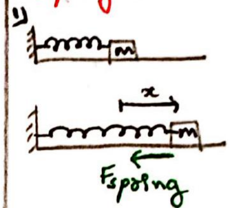
2) Find accn by $\boxed{\tau = I\alpha}$

3) We get $\boxed{\alpha = -(\text{const.})\theta}$ ↳ $\omega^2$

4) Since, $\boxed{T = 2\pi/\omega}$

## # Spring block & Simple Pendullum :

1)



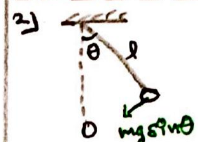

$F = -Kx$

$ma = -Kx$

$\boxed{a = -\left(\frac{K}{m}\right)x}$

⇒ $\boxed{T = 2\pi\sqrt{\frac{m}{K}}}$

2)




$\tau = -mg\sin\theta \times l$

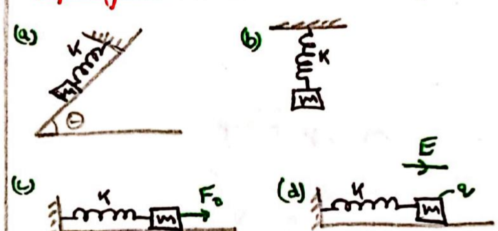
$I\alpha = -mgl\sin\theta$

$ml^2\alpha = -mgl\,\theta \quad (\theta \lll 1)$

$\boxed{\alpha = -(g/l)\theta}$

⇒ $\boxed{T = 2\pi\sqrt{\frac{l}{g}}}$

## # Spring Block : (under const. force)




In above all cases,

(i) Only mean position changes

(ii) Time period is same for all

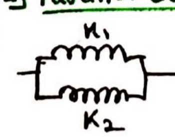
$\boxed{T = 2\pi\sqrt{\frac{m}{K}}}$

## # Combination of Springs :

1) **Series Combination ⇒**

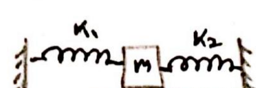

$\boxed{K_{eq} = \frac{K_1 K_2}{K_1 + K_2}}$

2) **Parallel Combination ⇒**




$\boxed{K_{eq} = K_1 + K_2}$

Eg 1.




$T = 2\pi\sqrt{\frac{m}{K_1 + K_2}}$

Eg 2.


$K_{eq} = \frac{2K}{2} + 2K = 3K$

$T = 2\pi\sqrt{\frac{m}{3K}}$

# Cutting of Spring:

→ For a spring, product of spring const & its length always remains constant.

$$\boxed{K \cdot l = \text{constant}}$$


$$K l = K_1 l_1 = K_2 l_2$$

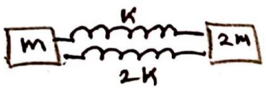
so, $\boxed{K_1 = \frac{Kl}{l_1}}$ & $\boxed{K_2 = \frac{Kl}{l_2}}$

# Two Block System:

Always used reduced mass instead of 'm'.

$$\boxed{\mu_{red} = \frac{m_1 m_2}{m_1 + m_2}}$$

Eg:

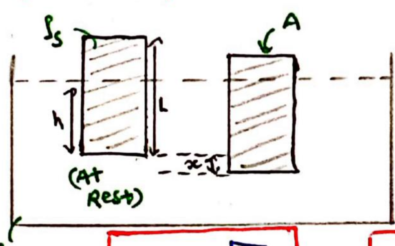



$$\mu_{red} = \frac{m \cdot 2m}{m + 2m} = \frac{2m}{3}$$

$$K_{eq} = K + 2K = 3K$$

so, $\boxed{T = 2\pi\sqrt{\frac{2m}{9K}}}$

# Block in Liquid: (When block is light)




$[\therefore m = \rho_L A h]$

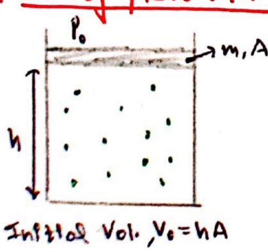
$\boxed{T = 2\pi\sqrt{\frac{h}{g}}}$ or $\boxed{T = 2\pi\sqrt{\frac{m}{A \rho_L g}}}$

At equilibrium ⇒

$\boxed{\rho_s / \rho_L = \frac{h}{L}}$ $\boxed{g_{eff} = g - \frac{F_b}{m}}$

$[F_b = \text{Mass displ.} \times g]$

# SHM of piston in cylinder:




Initial Vol. $V_0 = hA$

1) Piston given small displacement "Isothermally"

$$\boxed{T = 2\pi\sqrt{\frac{m V_0}{P_0 A^2}}}$$

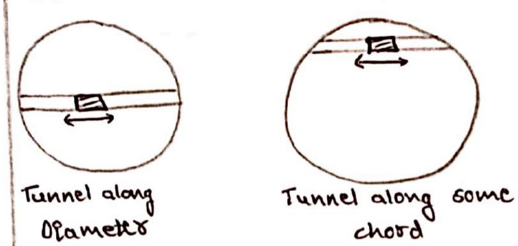
2) Piston given small displacement adiabatically

$$\boxed{T = 2\pi\sqrt{\frac{m V_0}{\gamma P_0 A^2}}}$$

Isothermal ke expression mein 'γ' add kardo

$[\gamma = C_P / C_V]$

# SHM in tunnel in a planet:



Tunnel along Diameter

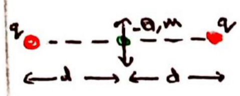
Tunnel along some chord

Expression is same in both cases,

$$\boxed{T = 2\pi\sqrt{\frac{R}{g}}} \quad \left(g = \frac{GM}{R^2}\right)$$

# SHM of charge:

(a) Vertical SHM ⇒

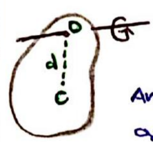



$$\boxed{T = 2\pi\sqrt{\frac{md^3}{2KQq}}}$$

(b) Horizontal SHM ⇒


$$\boxed{T = 2\pi\sqrt{\frac{md^3}{4KQq}}}$$

# Physical Pendulum:




$$\boxed{T = 2\pi\sqrt{\frac{I}{mgd}}}$$

Any rigid body can behave as a physical pendulum.

Here,

$I$ = MOI of body about O

$d$ = Distance b/w O & C (COM)

Eg:

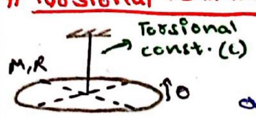
(Ring)

$$I = MR^2 + MR^2 = 2MR^2$$

$$d = R$$

$$I = 2\pi\sqrt{\frac{2MR^2}{MgR}} = 2\pi\sqrt{\frac{2R}{g}}$$

# Torsional Pendulum:




Torque is generated in opp. dirn of angular displacement

$$\tau = -C\theta$$

$$\Rightarrow I\alpha = -C\theta$$

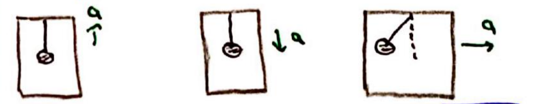
$$\alpha = -\underbrace{\left(\frac{C}{I}\right)}_{\omega^2}\theta$$

$$\boxed{T = 2\pi\sqrt{\frac{I}{C}}}$$

# SHM of dipole in field:


$\boxed{T = 2\pi\sqrt{\frac{I}{pE}}}$ $\boxed{T = 2\pi\sqrt{\frac{I}{MB}}}$

# T of simple pendulum:



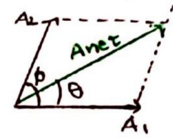

$g_{eff} = a + g$ $g_{eff} = g - a$ $g_{eff} = \sqrt{a^2 + g^2}$

$$\boxed{T = 2\pi\sqrt{\frac{L}{g_{eff}}}}$$

# Superposition of SHM:

⊙ Jis SHM ki eqn ko reference lo, uske pure phase mein θ add karna

⊙ Here, reference is $A_1$ SHM.




$y = A_1 \sin\omega t + A_2 \sin(\omega t + \phi)$

$\Rightarrow \boxed{y = A_{net}\sin(\omega t + \theta)}$

$$\boxed{A_{net} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\phi}}$$

$$\boxed{\theta = \tan^{-1}\left(\frac{A_2\sin\phi}{A_1 + A_2\cos\phi}\right)}$$

Eg: $y = 3\sin(\omega t + 30°) + 4\sin(\omega t + 120°)$

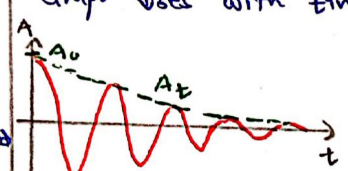
Soln Taking $A_1$ as reference,

$$y = 5\sin(\omega t + 30° + \theta)$$

$$\theta = \tan^{-1}(4/3)$$

# Damped Oscillation:

→ When a particle executes SHM in a medium other than air, its amp. ↓ses with time.




$$\boxed{y = A(t)\cos(\omega t + \phi)}$$

Where,

$\boxed{A(t) = A_0 e^{-\frac{bt}{2m}}}$ [b = Damping const. Depends on med.]

$$\boxed{\omega' = \sqrt{\frac{K}{m} - \frac{b^2}{4m^2}}}$$

# Simple Harmonic Motion - 2

## # Taylor's Theorem :

→ Used to solve SHM equations

→ Find the expression of restoring force responsible for SHM.

→ Then,

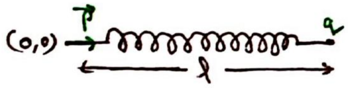
$$\boxed{F(x) = 0 + x \cdot F'(0)} \left\{ \text{Neglect higher order terms} \right.$$

So, $ma = x \cdot F'(0)$

$$a = x\left(\frac{F'(0)}{m}\right)$$

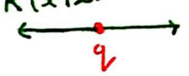
$$\Rightarrow \boxed{T = 2\pi\sqrt{\frac{m}{F'(0)}}}$$

Eg:




Find expression for time period of SHM of spring if given figure is of eqm state. (Natural length=0)

Soln Give small displacement $x$ to charge $q$ in right.




$+\frac{2pKq}{(l+x)^3}$

So, $F(x) = R(l+x) - \frac{2pKq}{(l+x)^3}$

$$F'(x) = R + 3\frac{(2pKq)}{(l+x)^4}$$

$$F'(0) = R + \frac{6pKq}{l^4}$$

Now, at eqm condition

$$Rl = \frac{2pKq}{l^3} \Rightarrow R = \frac{2pKq}{l^4}$$

So, $\boxed{\frac{6pKq}{l^4} = 3R}$

So, $F'(0) = R + 3R \Rightarrow 4R$

Thus, $\boxed{T = 2\pi\sqrt{\frac{m}{4R}}}$ Ans

## # Lissajous Figures :

→ When two SHM of ⊥ar dirn but same frequency are superimposed.

$x = A_1 \sin\omega t \qquad y = A_2 \sin(\omega t + \phi)$

[Eliminate t to get relation b/w x & y which will give eqn of path of particle.]

∴ $x = A_1 \sin\omega t$

$\Rightarrow \sin\omega t = \frac{x}{A_1} \qquad \cos\omega t = \sqrt{1 - \frac{x^2}{A_1^2}}$

Now, $y = A_2 \sin(\omega t + \phi)$

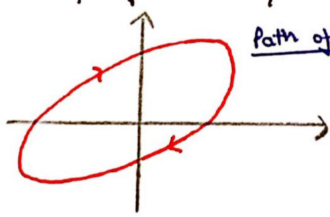
$$y = A_2(\sin\omega t\cos\phi + \cos\omega t \sin\phi)$$

$$y = A_2\left(\frac{x}{A_1}\cos\phi + \sqrt{1 - \frac{x^2}{A_1^2}}\sin\phi\right)$$

$$\Rightarrow \boxed{\frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} - \frac{2xy\cos\phi}{A_1 A_2} = \sin^2\phi}$$

↳ It is eqn of an oblique ellipse

Path of particle

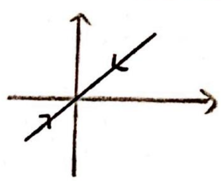


(i) If $\phi = 0°$

Then, $\frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} - \frac{2xy}{A_1A_2} = 0$

$$\Rightarrow \left(\frac{x}{A_1} - \frac{y}{A_2}\right)^2 = 0$$

$$\Rightarrow \boxed{y = \frac{A_2}{A_1}x}$$



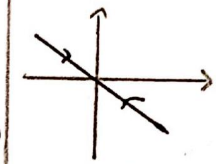
straight line with +ve slope

(ii) If $\phi = \pi$

Then, $\frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} + \frac{2xy}{A_1A_2} = 0$

$$\Rightarrow \left(\frac{x}{A_1} + \frac{y}{A_2}\right)^2 = 0$$

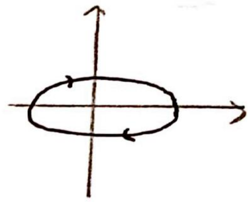
$$\Rightarrow \boxed{y = -\frac{A_2}{A_1}x}$$



straight line with -ve slope

(iii) If $\phi = \pi/2$

Then, $\boxed{\frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} = 1}$

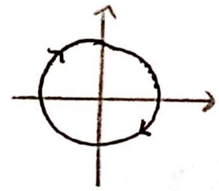


Standard Ellipse

(iv) If $\phi = \pi/2$ & $A_1 = A_2$

Then, $\frac{x^2}{A^2} + \frac{y^2}{A^2} = 1$

$$\Rightarrow \boxed{x^2 + y^2 = A^2}$$



Circle of radius 'A'

## # When ω is also different :

→ When ω is diff., then relation b/w both ω would be given. Use that to derive relation b/w x & y.

Eg: $x = A\sin(at + \delta)$

$y = B\sin(bt)$

Find Lissajous figure if $A = B$, $a = 2b$ & $\delta = \pi/2$.

Soln $x = A\sin(2bt + \pi/2)$

$$\Rightarrow \boxed{\frac{x}{A} = \cos 2bt}$$

Now, $y = A\sin bt$

Since, $\left[\sin^2\theta = \frac{1 - \cos 2\theta}{2}\right]$